WEEKLY TECHNICAL REPORT

Week 3 | Endpoint Logging & Network Detection

DG Interns Hub • Cyber Security Internship

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Evidence basis: provided Week 3, screenshots and local.rules; activity descriptions checked against supplied screenshots. Course completion/progress based on learner report and supplied training pages.

Executive Summary

Week 3 focused on endpoint and network detection in an isolated VirtualBox lab. I installed and used Sysmon on Windows 11, auditd on Ubuntu, and Snort 2.9.20 to capture security events. I authored four Snort rules and tested ICMP, SSH, HTTP and thresholded SYN activity. Four TryHackMe rooms and five LetsDefend courses were reported completed; the sixth LetsDefend Linux incident-response course had a VIP+ restriction on its interactive lab. I documented the limitation rather than submitting unverified answers.

Objectives & completion matrix

Required component	Evidence-based status	Proof / caveat
Theory notes – 10	Included	Ten separate Word documents in Notes/.
TryHackMe – 4 rooms	Learner-reported complete	Four room screenshots; no task-by-task transcripts.
LetsDefend – 6	5 reported complete; 1 partial	Five course screenshots and VIP+ popup.
Sysmon events	Demonstrated	Event IDs 1, 3, 11 visible in screenshots.
Linux auditd	Demonstrated	EXECVE apt install nano under key nano.
Snort & T1046	Demonstrated	Four local rules, config test, four alert tests.
Cheat sheet – 15+	Included	16-entry separate document and CSV.
LinkedIn posts	Not published	Both posts must be published manually; draft TXT files excluded as requested.

1. Lab topology and scope

The test network used VirtualBox host-only addressing (192.168.56.0/24). Snort ran on Ubuntu 192.168.56.101 using interface enp0s8; Kali 192.168.56.102 generated permitted scan/test traffic; Windows 11 showed source address 192.168.56.104 in Sysmon Event ID 3. All detection tests were contained within the personal lab.

System	Address in Week 3	Role
Ubuntu	192.168.56.101	auditd, Snort, target web/SSH services
Kali	192.168.56.102	Authorized ICMP, curl, nc and nmap tests
Windows 11	192.168.56.104	Sysmon and test client

2. Windows endpoint telemetry – Sysmon

Sysmon was installed with the SwiftOnSecurity config and verified in Event Viewer → Applications and Services Logs → Microsoft → Windows → Sysmon → Operational. Controlled activity produced events suitable for SOC inspection.

Event ID	Observed fields	Why it matters
1 / Process Create	Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe; ProcessId 7772; UtcTime 2026-11-03 05:48:42.375	Supports parent/child process triage and executable attribution.
3 / Network Connection	Protocol TCP; SourceIp 192.168.56.104:51420; DestinationIp 192.168.56.101:80	Maps outbound host connections to specific endpoints; this event alone is benign test traffic.
11 / File Create	Image powershell.exe; TargetFilename C:\Users\kesheth\Downloads\Week3_Sysmon_Test.txt; UtcTime 2026-11-03 06:26:23.270	Helps identify new files or dropped scripts in suspicious contexts.

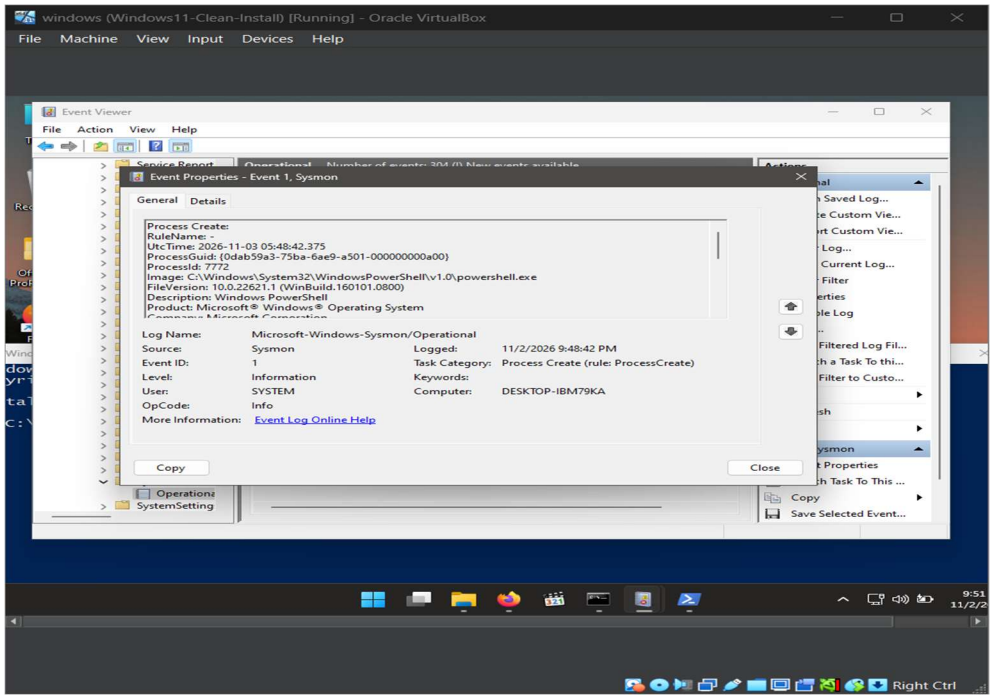


Figure 1 – Event ID 1: PowerShell execution.

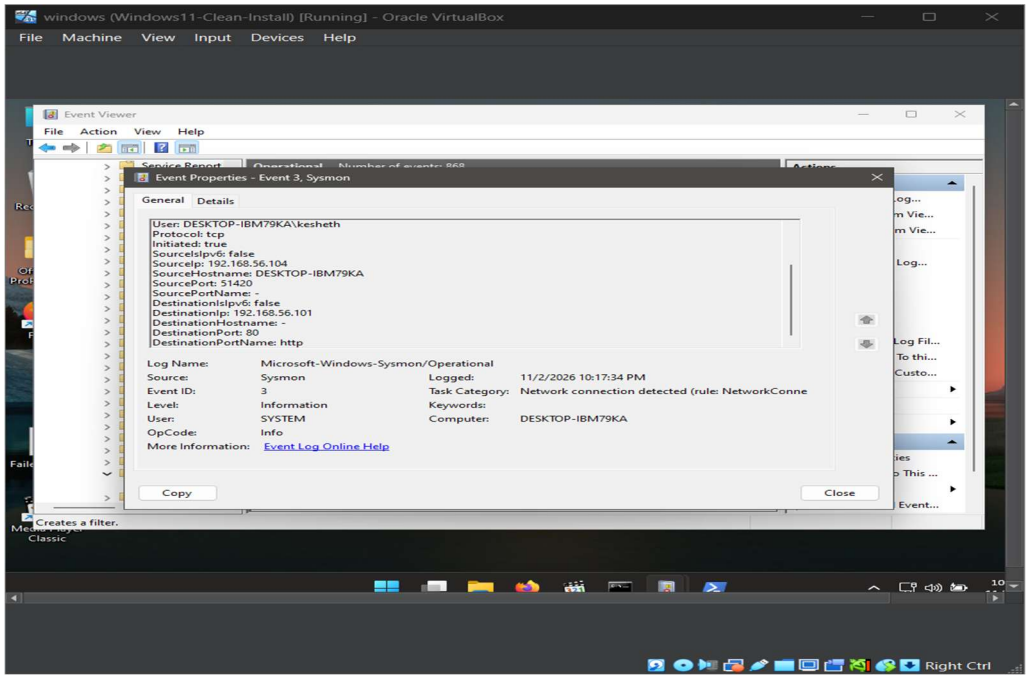


Figure 2 – Event ID 3: host-only TCP connection to Ubuntu port 80.

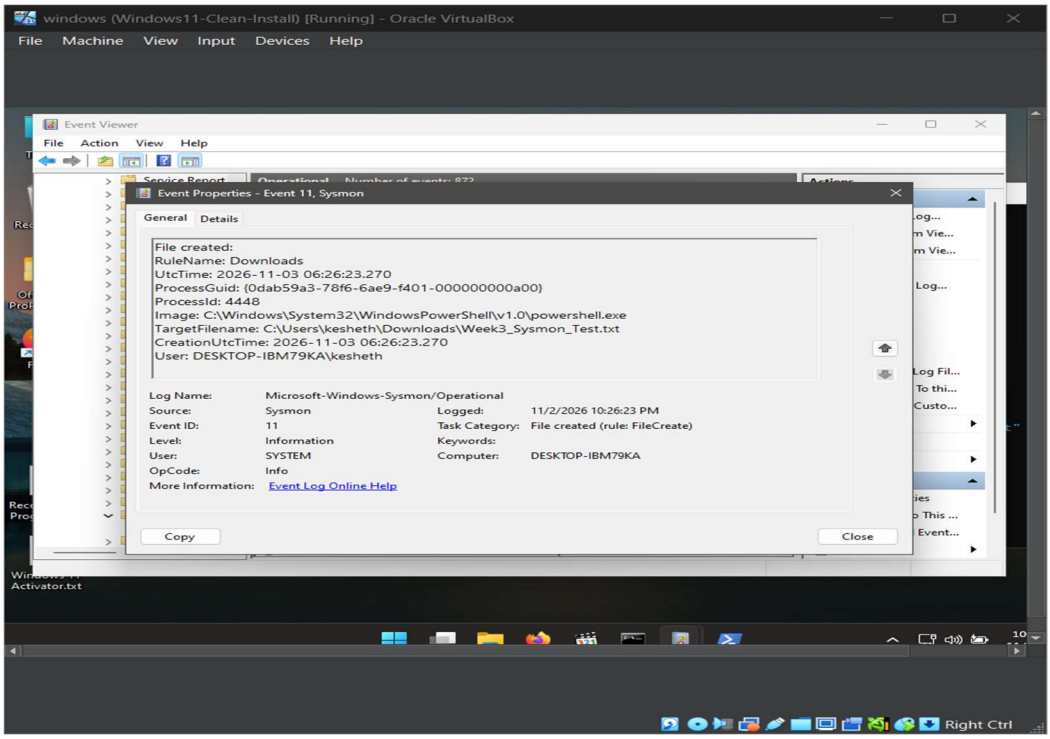


Figure 3 – Event ID 11: Downloads test-file creation.

3. Linux command auditing – auditd

On Ubuntu, auditd captured a controlled package-management action. A watch rule using key nano was applied to /usr/bin/apt; an apt install nano command was then located using ausearch -k nano -i. The screenshot shows PROCTITLE, PATH, EXECVE and SYSCALL record types, including a0=apt, a1=install, a2=nano, login/session and executable path fields.

```
sudo systemctl enable auditd --now
sudo auditctl -w /usr/bin/apt -p x -k nano
sudo apt install nano
sudo ausearch -k nano -i
```

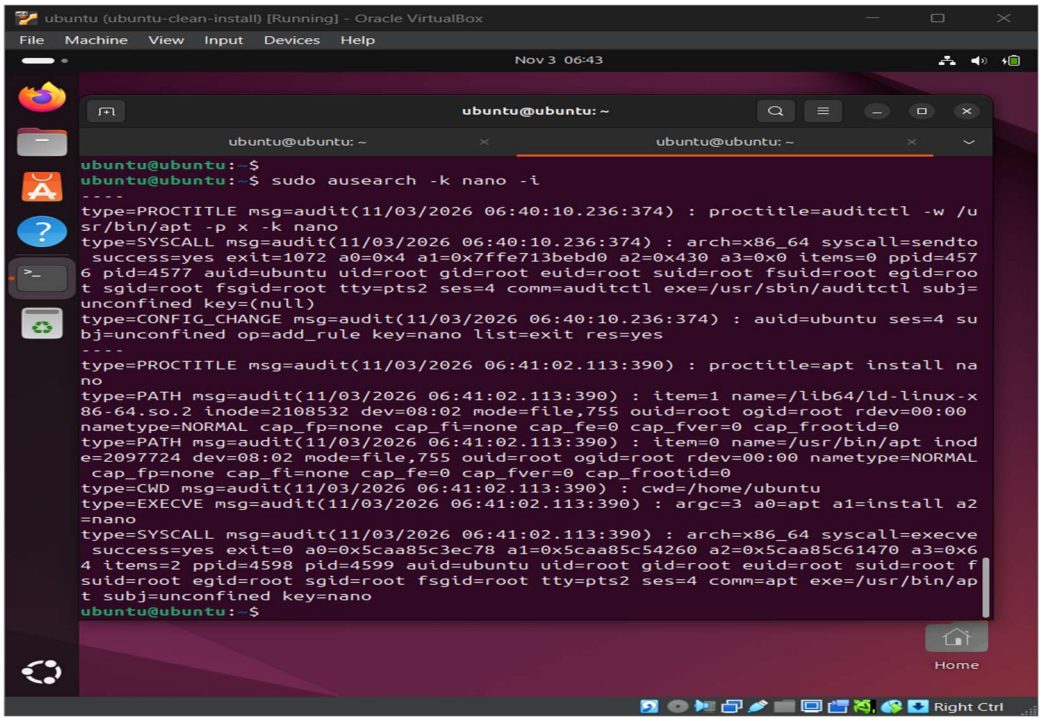


Figure 4 – The ausearch result contains an EXECVE record for apt install nano.

SOC relevance: the audit records tie an action to process arguments and actor details. The permissioned test command is an example of normal activity, not evidence of an intrusion.

4. Snort IDS and custom detection rules

Snort version 2.9.20 was used in console IDS mode on enp0s8. The HOME_NET variable was set to 192.168.56.0/24. Four local rules were included in Snort Rules/Week3_local.rules, with SIDs 1000001–1000004.

SID	Detection	Method / tested traffic
1000001	ICMP Ping Detected	Kali ping → Ubuntu; ICMP alert screenshot.
1000002	SSH Connection Attempt	Kali TCP/22 connection attempt → Ubuntu; SSH rule alert screenshot.
1000003	HTTP Traffic	Kali curl → Ubuntu:80; HTTP alert screenshot.
1000004	MITRE T1046 – Nmap SYN Port Scan Detected	Kali nmap -sS -T4 -p 1-1000; thresholded TCP SYN alert screenshot.

```
sudo snort -T -i enp0s8 -c /etc/snort/snort.conf
sudo snort -A console -q -c /etc/snort/snort.conf -i enp0s8
sudo nmap -sS -T4 -p 1-1000 192.168.56.101
```

Deviation and justification: unlike the guide's normal EXTERNAL_NET=!$HOME_NET example, EXTERNAL_NET=any was used in this isolated lab because Kali (192.168.56.102) is within HOME_NET (192.168.56.0/24). With !$HOME_NET, the externally directed rule would not match the authorized Kali SYN traffic. This lab-specific simplification broadens the matching scope; it should not be copied to a production deployment without reassessing rule direction and noise.

The T1046 rule detects bursts of SYN packets satisfying flags:S and count 20 / 1 second; it cannot attribute traffic to the Nmap binary and may flag benign high-rate connections. ATT&CK mapping is analyst contextualization, not proof of intent.

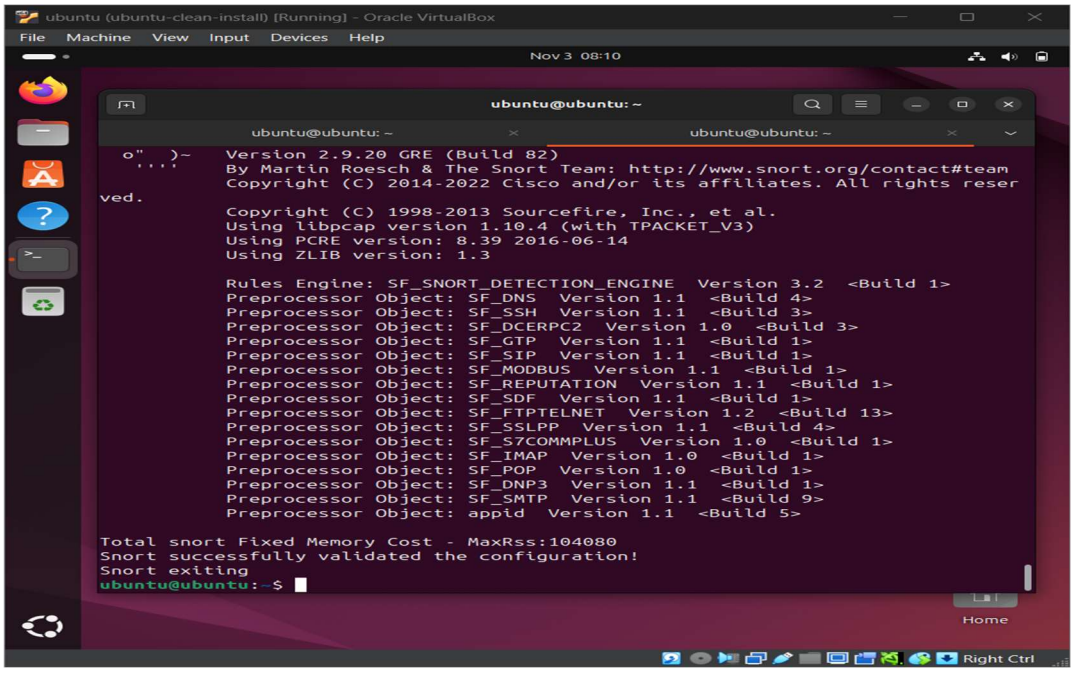


Figure 5 – Snort successfully validates the four configured local rules.

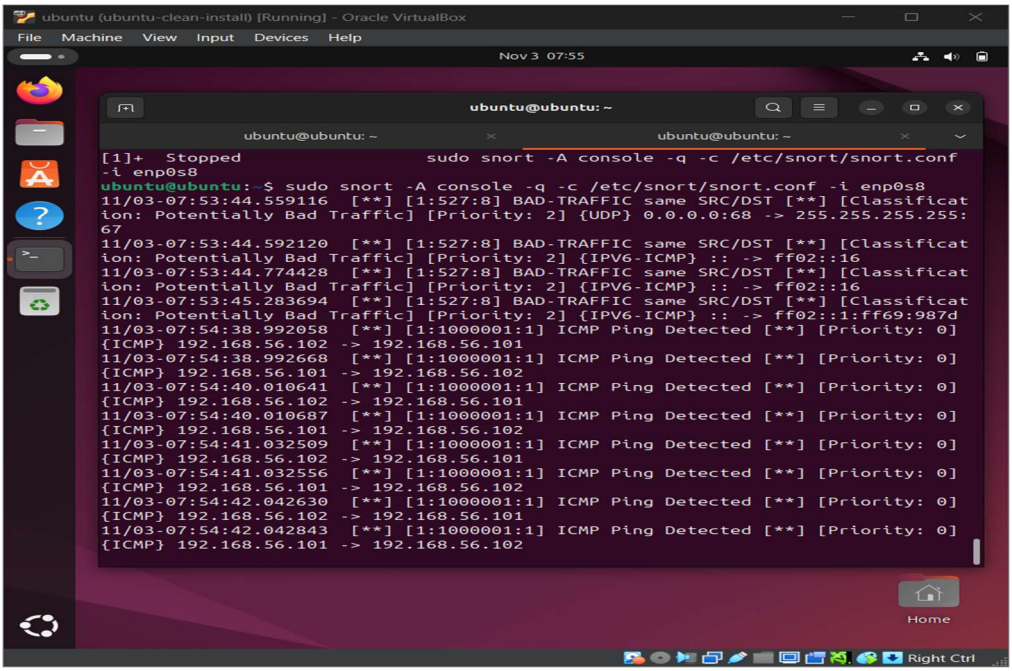


Figure 6 – ICMP alert in console output.

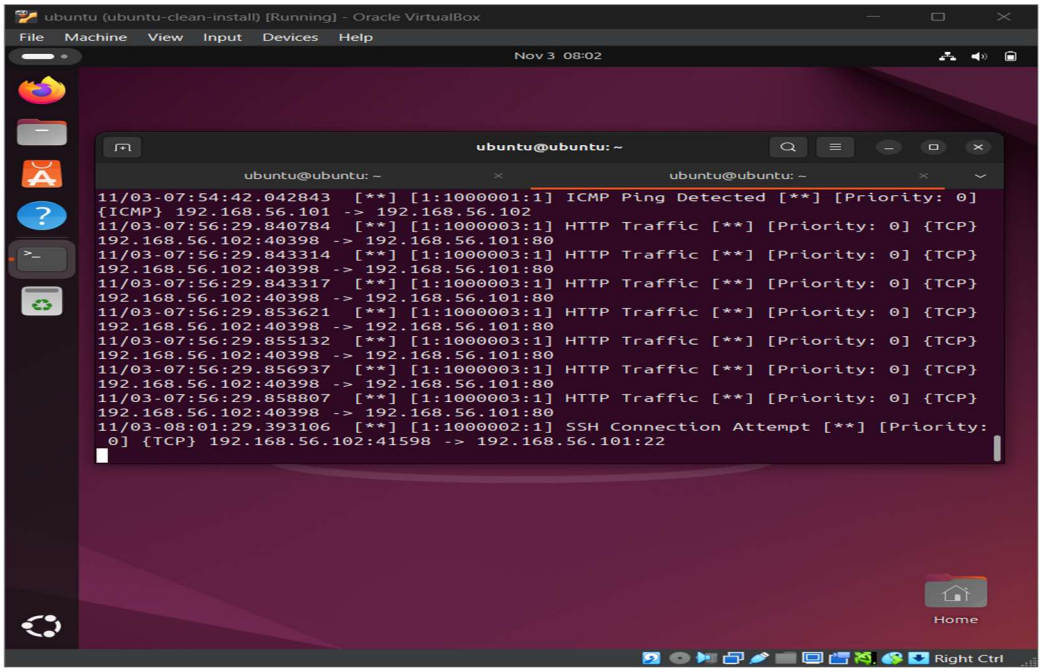


Figure 7 – SSH connection attempt alert in console output.

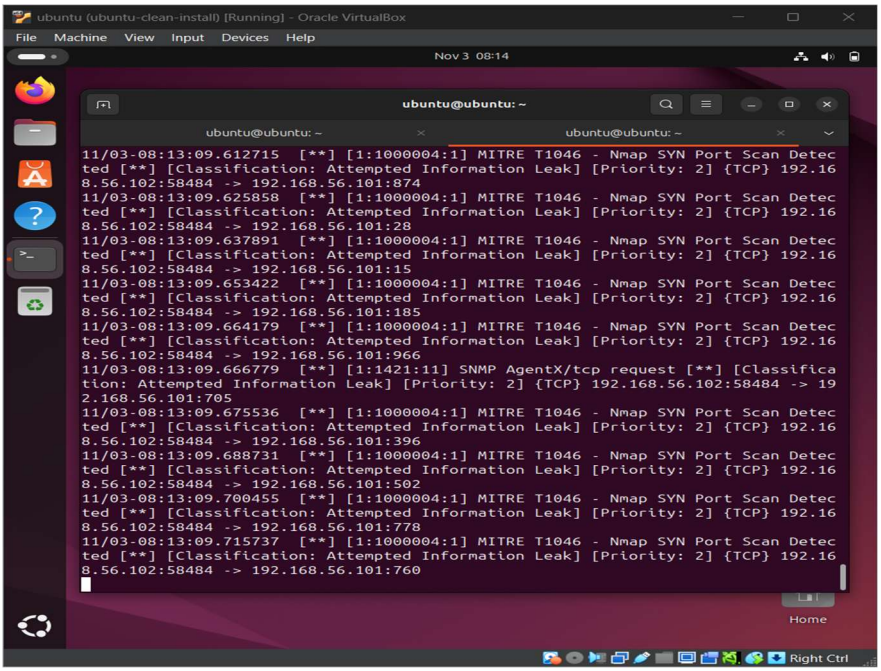


Figure 8 – Alert from SID 1000004 during the Nmap SYN test.

5. Training labs and integrity of evidence

TryHackMe: Linux Logging for SOC, Intro to Logs, Intro to Log Analysis, and Snort were reported completed. Four platform screenshots are provided. LetsDefend: How to Investigate a SIEM Alert, Security Solutions, Network Log Analysis, SIEM 101 and Incident Management were reported completed; screenshots show the training pages. One writeup per required room/course is included. The screenshots are not represented as full command-by-command transcripts.

Incident Response on Linux: the course page was accessible, but clicking the Lab Environment Connect control showed the VIP+ plan comparison and $39.99/month price. The interactive lab investigation and several questions were not completed; attempts to infer answers from public writeups were not used as verified evidence.

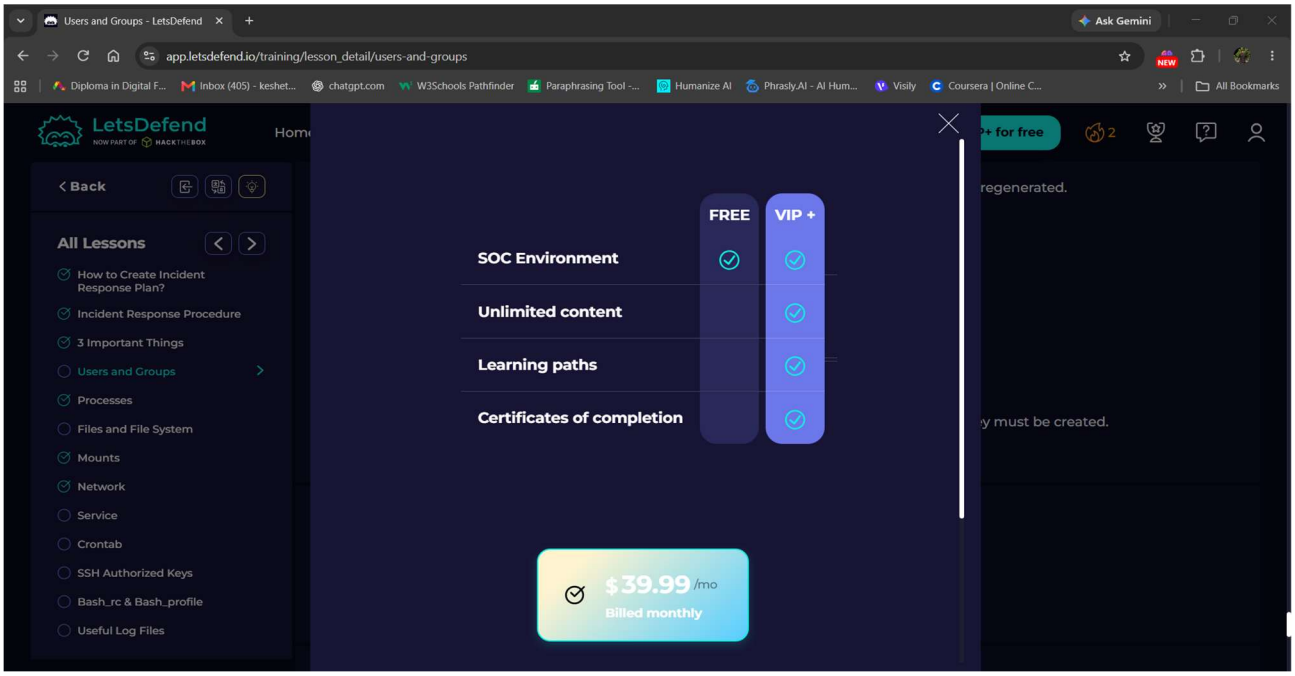


Figure 9 – VIP+ subscription dialog triggered when connecting to the lab.

6. Challenges, diagnostics and resolution

Observed problem	Diagnostic action / evidence	Resolution or limitation
Windows user creation initially returned System error 5	Insufficient privilege when creating test user	Used elevated Administrator PowerShell; Sysmon user-creation process screenshot retained.
Ubuntu VirtualBox clipboard / guest integration mismatch	Guest Additions / VBoxClient issue; initial package name typed incorrectly	Installed correct virtualbox-guest-x11 package, restarted desktop and used shared folder Week3Share to transfer local.rules.
Snort interface naming differed from examples	Verified active NIC with ip a	Used Ubuntu host-only interface enp0s8, not eth0 or enp0s3.
Snort HOME_NET / EXTERNAL_NET scoping	Kali test host is inside HOME_NET	Set EXTERNAL_NET any for a controlled test, documented security trade-off.
LetsDefend interactive Linux lab VIP+	Connect opened subscription comparison	Saved evidence, left lab questions incomplete; continued using free materials.

Troubleshooting items above combine previous hands-on notes with the evidence provided in the ZIP; screenshots primarily prove the eventual working outcome rather than every failed step. Commands that were not screenshot-captured as errors are not misrepresented as directly evidenced.

7. Key lessons & recommended improvements

- Endpoint logs give context beyond basic OS errors: process image, user, arguments, file path and network destination.
- Valid detections require context; an expected internal scan is not an incident.
- Rule variables and topology matter. A correct rule can miss expected test traffic if source zones are scoped incorrectly.
- Audit logs should record actor, time, executable, result and arguments for reliable investigation.
- Next lab: confirm Sysmon Event IDs 13 and 22 and compare false-positive rates of the SYN threshold. These were not demonstrated in the supplied screenshots.

8. Submission audit and evidence references

The final package contains Notes (10 independent answers), Writeups (4 TryHackMe + 6 LetsDefend + 4 hands-on), Findings (original screenshots), Cheat Sheet (16 entries), Snort Rules (original four-rule file), Word Report, PDF Report, Presentation Report and README. The four extra root files requested for removal are not included.

The two LinkedIn posts are required external actions; draft TXT files were removed from this submission ZIP as requested. Publication on LinkedIn and submission of a shared Google Drive link are not claimed complete.

Evidence reference: original screenshot files are included under Findings/, arranged by platform (Sysmon, auditd, Snort, TryHackMe and LetsDefend). No separate evidence inventory is included.